Omkar Mayekar

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Experience

Graduate Research Assistant, *Clemson University* 02/2025 – 05/2026 Clemson, USA

- Extended and evaluated a geometry-guided latent diffusion pipeline initiated by a senior PhD researcher, benchmarking it against state-of-the-art view synthesis methods on off-road simulation and KITTI datasets; authored the paper and produced the pipeline/comparison figures and evaluation tables.
- Improved 3D Gaussian Splatting reconstruction for real-world scenes by adding robust camera-intrinsic-aware data loading, reducing reconstruction artifacts and improving spatial scene fidelity.

Graduate Teaching Assistant, *Clemson University* 08/2025 – 12/2025 Clemson, USA

- Guided students in setting up and using the LeRobot framework for robot learning and manipulation.
- Provided code templates, pipelines, and technical support for recording datasets within simulation environments.

Education

Master of Science in Computer Science, *Clemson University* 2026 Clemson, USA

Bachelor of Technology in Computer Engineering, *University of Mumbai* 2022 Mumbai, India

Project

VIEW TRANSLATION: GEOMETRY-GUIDED LATENT DIFFUSION FOR CROSS-VIEW IMAGE SYNTHESIS ON GROUND VEHICLES, *Advanced Imaging Lab*

- Extended and experimentally evaluated a geometry-guided latent diffusion pipeline originally initiated by a senior PhD researcher, conducting systematic comparisons against state-of-the-art view synthesis methods on off-road simulation and KITTI datasets.
- Authored the full research manuscript and led the technical communication of the work, including designing the pipeline and comparison figures, constructing evaluation tables, analyzing benchmark results, and documenting the experimental findings.

Generalizable Neural Radiance Field Super Resolution, *Advanced Imaging Lab*

- Implemented, analyzed, and benchmarked state-of-the-art 3D reconstruction and neural rendering methods including NeRF-SR, 3D Gaussian Splatting, Instant-NGP, and TensoRF to evaluate reconstruction quality and compare the performance of our super-resolution method on real-world and synthetic datasets.
- Evaluated generalizable NeRF approaches including GeFu and GDB-NeRF to understand their cross-scene generalization capabilities and benchmark their performance against our method across real-world and synthetic scenes.

3D Collision Detection in Robot Task Manipulation, *Advanced Imaging Lab*

- Implemented state-of-the-art policies including ACT, SmolVLA, and GrootN1.5 on the SO101 real-world robot arm and studied their limitations.
- Implemented VGGT and Depth Anything 3 for real-world 3D scene understanding, enabling robot manipulation policies to use spatial context for collision-aware action planning.

Publications

- **View Translation: Geometry-Guided Latent Diffusion for Cross-View Image Synthesis of Ground Vehicles** — Published paper.
- **Generalizable Neural Radiance Field Super-Resolution** — arXiv paper.

Skills

Programming
Python, C++, XML, PyTorch, OpenCV

Software
MuJoCo, Isaac Sim, LeRobot, COLMAP, Nerfstudio, GitHub Copilot

OS
Windows, Linux, Palmetto HPC Cluster (SLURM)